

Homework 8

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1 if $a \mid b$ and $a \mid c$ then $a \mid (b + c)$

$$\text{let } b = ax, x \in \mathbb{Z} \tag{1}$$

$$\text{let } c = ay, y \in \mathbb{Z} \tag{2}$$

$$b + c = ax + ay \tag{3}$$

$$b + c = a(x + y) \tag{4}$$

$$\text{let } p = x + y \tag{5}$$

$$p \in \mathbb{Z} \tag{6}$$

$$\therefore p \in \mathbb{Z} \tag{7}$$

$$b + c = ap \tag{8}$$

$$\therefore \text{if } a \mid b \text{ and } a \mid c \text{ then } a \mid (b + c)$$

2 if $a \mid b$ then $a \mid bc$, for all integer c

$$\text{let } b = ax, x \in \mathbb{Z} \tag{9}$$

$$bc = axc, c \in \mathbb{Z} \tag{10}$$

$$bc = a(xc) \tag{11}$$

$$\tag{12}$$

$$\therefore \text{if } a \mid b \text{ then } a \mid bc, \text{ for all integer } c$$

3 if $d \mid n$ and $n \mid d$ then $d = \pm n$

$$\text{let } d = nk_1, k_1 \in \mathbb{Z} \quad (13)$$

$$\text{let } n = dk_2, k_2 \in \mathbb{Z} \quad (14)$$

$$d = dk_1k_2 \quad (15)$$

$$1 = k_1k_2 \quad (16)$$

$$k_1 = \frac{1}{k_2} \quad (17)$$

$$\text{from } k_1 \in \mathbb{Z} \text{ and } k_2 \in \mathbb{Z} \quad (18)$$

$$\therefore k_2 \in \{-1, 1\} \quad (19)$$

$$\therefore k_1 \in \{-1, 1\} \quad (20)$$

$$d = \pm n \quad (21)$$

$$n = \pm d \quad (22)$$

$$(23)$$

$\therefore$ if $d \mid n$ and $n \mid d$ then $d = \pm n$